Kiera (Ngoc Thuy An) Tran

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Google Scholar | LinkedIn

Education

Ph.D. in Earth and Atmospheric Sciences — Anticipated 07/2027
Georgia Institute of Technology, Atlanta, GA
Advisor: Dr. Winnie (Wing Yin) Chu | Group's Website

B.S. in Earth and Atmospheric Sciences (Highest Honor) — May 2022
Georgia Institute of Technology, Atlanta, GA — GPA: 3.94/4.00

Publications

Tran, K., Chu, W., Yang, D., Dawson, E., Padman, L., Cordero, I., Tinto, K., & Bell, R. (submitted). Radar attenuation constrains spatially variable basal melt rates across Ross Ice Shelf. *Sponsored by NASA FINESST Fellowship*.

Field Experience

NASA Student Airborne Research Program, Houston & Richmond, USA — May – Jul 2026
Mentored 6 undergraduate students to develop research on land remote sensing
Airborne surveyed atmospheric compositions and air quality over Texas using UAVSAR, GCAS, AWP, AVIRIS, and HSRL; in addition to ground-truthing trips

Patagonia Icefield Research Program, Torres de Paine, Chile — Nov 2023 – Jun 2024
Learned about geophysical methods and applications in glaciology and geomorphology
Self-led a ground-penetrating radar (GPR) survey

Observational Seismology Field Trip, South Carolina, USA — 2022 – 2023
Deployed 86 passive SmartSolo seismometers and a broadband sensor to record the sequences of earthquakes happening in South Carolina.

Geophysical Method Field Trip, Harvard Hill, California, USA — 19 – 26 Mar. 2022
Applied geophysical instrumentation (magnetometer, GPR, BadElf Flex – GPS, and seismic geophones) to detect active/inactive faults in the Harvard Hill of the Mojave Desert, a part of the tectonically active San Andreas Fault.

Research Experience

Graduate Research Assistant, Georgia Institute of Technology, Atlanta, GA

An airborne study on West Antarctic ice-shelf basal channels — 2022 – present
Applied machine learning and optimization techniques to airborne radar data for detecting basal channel formation and tracking its evolution in West Antarctica.
Collaborator: Dr. Karen Alley - University of Manitoba, Winnipeg, Canada

Basal Melting Observations of Ross Ice Shelf (RIS) in Antarctica — 2022 – 2026
Developed algorithms for estimating ice-shelf melt rates using radar attenuation and temperature modeling.

Modeling and predicting the impact of seawater intrusion Summer 2024

Simulated 100-year seawater intrusion impacts with ISSM modeling framework.

Collaborator: Dr. Frank Pattyn - Université libre de Bruxelles, Brussels, Belgium

Ice thickness measurement in Grey Glacier Summer 2023

Processed radar signals to estimate ice thickness and conducted field surveys for validation.

Collaborator: Dr. Camilo Rada - University of Magallanes, Punta Arenas, Chile

Bistatic Radar Development 2022 – 2023

Built a multi-transmitter/receiver radar system, including signal processing workflows for 3D subsurface temperature profiling

Undergraduate Research Assistant, Georgia Institute of Technology, Atlanta, GA

Forty-year of Ross Ice Shelf Change 2021 – 2022

Analyzed 40+ years of radar datasets of Ross Ice Shelf using **computational methods** for long-term stability observation & modeling.

Biological, ecological, and environmental flow applications Spring 2021

Conducted turbulence analysis of aquatic organism response using simulation and data-driven methods.

Conference Abstracts

[14] Fellows, M.*, Tran, K., Osmanoglu, B., & Dechow, J. (2026). Using SAR to Identify Tornado Tracks from the March 14th, 2025 Midwestern US Outbreak, *American Meteorological Society Annual Meeting*.

[13] **Tran, K.**, Chu, W., Yang, D., Dawson, E., Padman, L., Cordero, I., Schroeder, D., & Bell, R. (2026). Airborne Radar Observations of Basal Melt Rates Linked to Cavity Ocean Circulation at Ross Ice Shelf, *Ocean Science Meeting*.

[12] **Tran, K.**, Chu, W., & Alley, K. (2024). Airborne Radar Observations of West Antarctic Ice-Shelf Basal Channels, *AGU Fall Meeting*.

[11] **Tran, K.**, Chu, W., Yang, D., Padman, L., Dawson, E., Cordero, I., & Schroeder, D. (2024). Understanding Ross Ice Shelf Stability from Radar Observed Basal Melt Rates, *AGU Fall Meeting - Oral Talk*.

[10] Verboncoeur, H., Hills, B., Siegfried, M., Holschuh, N., Tarzona, A., **Tran, K**., Schroeder, D., & Chu, W. (2024). Internal and Basal Processes Driving Multi-Decadal Stress-Balance Changes in the Crary Ice Rise Region, West Antarctica, *AGU Fall Meeting*.

[9] **Tran, K.**, Chu, W., Padman, L., Schroeder, D., Cordero, I., & Tarzona, A. (2023). Integrating Basal Melt Rates from ROSETTA-Ice Airborne Sounding Radar Survey in Ross Ice Shelf, Antarctica, *AGU Fall Meeting*.

[8] Tarzona, A., Chu, W., Verboncoeur, H., Siegfried, M., Schroeder, D., Altaweel, A., Amaro, B., & **Tran, K.** (2023). Improved vertical Calibration of the Historical SPRI-NSF-TUD Airborne Radar Echo Sounding Ice thickness Measurements at Ross Ice Shelf, Antarctica, *AGU Fall Meeting*.

[7] **Tran, K.**, Chu, W., Padman, L., Schroeder, D., Cordero, I., & Tarzona, A. (2023). Integrating Basal Melt Rates of Ross Ice Shelf from ROSETTA-Ice Airborne Sounding Radar Survey and NASA/UCI Ice Sheet System Model, *WAIS Workshop*.

[6] **Tran, K.**, Chu, W., Tarzona, A., Schroeder, D., & Cordero, I. (2022). Basal Melting of Ross Ice Shelf based on ROSETTA-Ice Radio-echo Sounding Observations, *AGU Fall Meeting*.

[5] Tarzona, A., Chu, W., Verboncoeur, H., Siegfried, M., Schroeder, D., Combs, L., Prabu, A., Altaweel, A., & **Tran, K.** (2022). Geographical Repositioning Efforts and Vertical Calibration of Z-scopes from SPRI-NSF-TUD surveys at Ross Ice Shelf, Antarctica, *AGU Fall Meeting*.

[4] **Tran, K.**, Chu, W., Tarzona, A., Schroeder, D., & Cordero, I. (2022). ROSETTA-Ice Observations on Basal Melting of Ross Ice Shelf using Airborne Radio-echo Sounding Radar, *WAIS Workshop*.

[3] Tarzona, A., Chu, W., Verboncoeur, H., Siegfried, M., Schroeder, D., Combs, L., Altaweel, A., Prabu, A., & **Tran, K.** (2022). Archival airborne radio-echo sounding data geographical repositioning and calibration progress at Ross Ice Shelf, Antarctica, *WAIS Workshop*.

[2] Tarzona, A., Chu, W., **Tran, K.**, Teisberg, T., & Dawson, E. (2021). Four-decades of Ross Ice Shelf Change: Part 2. Comparison using archival SPRI-NSF-TUD and modern ROSETTA-Ice and NASA/NSF IceBridge radio-echo sounding data, *AGU Fall Meeting*.

[1] **Tran, K.**, Chu, W., Tarzona, A., Teisberg, T., & Dawson, E. (2021). Four-decades of Ross Ice Shelf Change: Part 1. Modern ice shelf basal conditions based on ROSETTA-Ice and NASA-/NSF IceBridge radio-echo sounding observations, *AGU Fall Meeting*.

Teaching & Mentoring Experience

Teaching

Polar STEAM

Research Scientist Fellow — 2026 – 2027

Georgia Institute of Technology, Atlanta, GA

Facilitator (Teams for Tech)

GT Leadership Education and Development (LEAD) — 2024 – 2025

Lecturer

GT 6000: Community, Mentorship, & Support — Fall 2024

Guest Lecture

EAS 3610: Introduction to Geophysics, 'Heat Flow' — 2023

Undergraduate/Graduate Teaching Assistant — 2022 – 2023

EAS 1600: Introduction to Environmental Science

EAS 1601: Habitable Planet

EAS 2600: Earth Processes

Vien Ngo Temple, Loganville, GA — 2023 – 2025

Language Instructor: Vietnamese

ABC-tutor, Atlanta, GA — 2020 – 2022

K-12 & College Prep Private Tutor & Mentor

University of North Georgia 2018 – 2020

STEM Tutor

Mentoring

NASA Student Airborne Research Program Summer 2026

Alissa Ferguson, Rice University

Catherine Pope, University of Vermont

Charlie Rosemond, Southern Utah University

Ella Boussy, University of Miami

Jackson Retelle, Embry-Riddle Aeronautical University

Morgan Fellows, Vermont State University Lyndon

Undergraduate Student Advising

Rowan Ray, GT Biology, 2024 – now

Daniel Park, GT Computer Science, 2024

Kieran Choi-Slattery, GT Aerospace Engineering, 2022 – 2023

Abdullah Altaweel, GT Computer Science, 2022 – 2023

Anh Nguyen, GT Industrial Engineering, 2022 – 2023

High School Student Advising

Huy Huynh, 2022 – 2023

Ben Huynh, 2021 – 2022 (*Now: Full-bright Undergraduate Student in Mechanical Engineering at UAH)*

Academia experiences

Administration & Leadership

ACC Advocacy Days at Capitol Hill, Washington DC 2024 – 2025

Advocated for increasing federal funding and more support and opportunities for international students

Georgia Tech Graduate Student Government Association

Worked with administrators from the President's Office to the department level to address concerns related to graduate students and build programs to support them, representing more than 30,000 on-campus and online graduate students

President 2024 – 2025

Executive Vice President 2023 – 2024

Vice President of Campus Services 2022 – 2023

Women in Geosciences, member 2022 – 2025

Graduate in Earth and Atmospheric Sciences, Secretary 2023 – 2024

Society of Exploration Geophysicists, Vice President 2022 – 2023

Guest Speaker

Virginia Commonwealth University, Richmond, Virginia May 2026

Title: *Machine Learning Applications in Earth and Environmental Sciences*

National Central University, Taoyuan, Taiwan Mar. 2026

Title: *Beneath the Surface: An Airborne Radar Investigation on Ice Shelf Basal Conditions and their Influences on the Antarctic Ice*

Du & Hoc Podcast*, 'Save the Planet'* 30 Oct. 2021

Academic Outreach

Georgia Tech, *CRIDC* 2023 – 2025

Georgia Tech, Geophysics Seminar, EAS department 2024

The Atlanta Children's Shelter, *'Invisible Ink & Chromatography'* Jul. 2023

Fort Valley State University*, 'A way to study ice'* Mar. 2023

Georgia Tech, Graduate Student Symposium, EAS department 2023

Spelman College, *Climate Sustainability Challenges & Opportunities* 2023

Stone Mountain Middle School*, 'Glaciers and Sea Level Rise'* 10 Nov. 2022

Middle school visits*, 'Glaciology'* Jun. 2022

Atlanta Science Festival*, 'Earthquake Machine'* 19 Mar. 2022

Georgia Tech*, EXPLORE Science and Math!* 2022

Volunteer

Science Olympiad @ Georgia Tech 2025

Chattahoochee Riverkeepers 2018 – present

Wikipedia Editor 2020 – 2021

Honors & Awards

NASA FINESST Fellowship ($150,000) 2023 – 2026

West Antarctic Ice Sheet Workshop

Travel Grant ($700) 2023

Georgia Institute of Technology, Atlanta, GA

EAS Travel Grant ($1,600) 2025

SGA Fund ($1450) 2024 – present

Center for Promoting Inclusion and Equity in the Sciences ($500) 2024

CRIDC - Poster Competition Winner ($1500) Feb 8, 2024

President's Fellowship ($22,000) 2022 – present

Quarter Century Award 2021 – 2022

Student of the Month Oct 2021

Dean's List 2020 – 2022

University of North Georgia, Dahlonega, GA

Third Place College Division Nengajo Contest 2020

President's Honor Roll 2019 – 2020

President's List 2017 – 2019

AlaMATYC Math Tournament, Tuscaloosa, AL

First Place Individual Calculus Competition ($200) 2017 & 2019

First Place Team Competition 2017

Relevant Courses

Global Cryosphere	Glaciers & Ice Sheet Dynamics
Ocean Dynamics	Sea Level Rise & Geocoastal Techniques
Oceanography	Physical & Chemical Oceanography
Deep Ocean	Paleoclimatology & Paleoceanography
Practical Computing & Math	Machine Learning in Environmental Sciences
Advanced Data Analysis	Land Remote Sensing
Introduction to Geophysics	Geophysical Field Methods
Physical Volcanology	Geodynamics

Technical skills

Coding languages:	Python, MATLAB
Libraries/Frameworks:	NumPy, Pandas, SciPy, scikit-learn
Tools:	Q-GIS, ArcGIS, Adobe Illustrator

Referees

Dr. Winnie Chu (Ph.D. primary advisor)
Assistant Professor
School of Earth and Atmospheric Sciences
Georgia Institute of Technology
Email: wchu38@gatech.edu

Dr. Karen Alley (Ph.D. co-advisor)
Associate Professor
Department of Environment and Geography
University of Manitoba
Email: karen.alley@umanitoba.ca

Cynthia Hall
Support Scientist
Early Career Research Program
NASA Earth Science Division
Email: cynthia.r.hall@nasa.gov